

Database Implementation Report - CYBR3410 - Grant Bryan

9/12/26

BLUF

This lab tasked me with implementing the ERD and the representative dataset. This lab also asked me to explain any changes to the ERD that were made during implementation. In addition to this, the lab asked me to verify that constraints and key controls were working as intended. I was successful in implementing my ERD design, along with adding more check constraints for security purposes. The representative dataset was also implemented successfully and used to verify my constraints and key controls.

Environment

The environment we will be working in for these decisions is our database on an Ubuntu 24.04 virtual machine mounted to our host machine, Fedora 7.1.12-200.fc44.x86_64. For this virtual machine, we are using VMware Workstation Pro 26H1, version 26.0.0.25388281, as our chosen hypervisor. For our database, we are using MySQL 8.4 LTS.

Schema Verification

The database was created and named `ngts_security_lab` as instructed. In addition to this, all tables have been created with configurations according to the ERD designed last week, with the exception of additional security features and the change of the `admin_boolean` to a less restrictive data type.

```
mysql> SHOW TABLES;
+-----+
| Tables_in_ngts_security_lab |
+-----+
| billing                      |
| customer                    |
| customer_devices             |
| department                   |
| employee                     |
| employee_compensation        |
| employee_tax_info            |
| technician_assignment         |
| ticket_junction_table        |
| tickets                      |
+-----+
10 rows in set (0.01 sec)

mysql> █
```

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Controls Analysis

Billing

```
+-----+
| billing | CREATE TABLE `billing` (
| `billing_id` int NOT NULL,
| `customer_id` int DEFAULT NULL,
| `customer_billing_address` varchar(255) DEFAULT NULL,
| `cardlast_four` varchar(4) DEFAULT NULL,
| PRIMARY KEY (`billing_id`),
| UNIQUE KEY `customer_id` (`customer_id`),
| CONSTRAINT `billing_ibfk_1` FOREIGN KEY (`customer_id`) REFERENCES `customer` (`customer_id`),
| CONSTRAINT `billing_chk_1` CHECK ((char_length(`cardlast_four`) = 4))
| ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+
```

Several controls have been added to ensure data integrity and key control. Customer_id foreign key has been added for increased security and data normalization. I have also added a check constraint to verify the card number last 4 is always 4 digits, due to VARCHAR accepting inputs less than 4 characters. I chose VARCHAR(255) as the data type for billing address to limit characters input in this field due to it being a relatively short phrase normally. This adds data integrity and makes sure space is being used correctly.

Customer Devices

```
+-----+
| customer_devices | CREATE TABLE `customer_devices` (
| `device_id` int NOT NULL,
| `customer_id` int DEFAULT NULL,
| `ip` varchar(45) DEFAULT NULL,
| `device_type` varchar(100) DEFAULT NULL,
| `host_name` varchar(100) DEFAULT NULL,
| `mac_address` varchar(17) DEFAULT NULL,
| PRIMARY KEY (`device_id`),
| UNIQUE KEY `mac_address` (`mac_address`),
| KEY `customer_id` (`customer_id`),
| CONSTRAINT `customer_devices_ibfk_1` FOREIGN KEY (`customer_id`) REFERENCES `customer` (`customer_id`)
| ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+
```

VARCHAR() was used extensively in this table for data integrity. IP has a hard limit of 45 characters to include IPv4 and IPv6 IP addresses. Device_type and host_name have been limited to 100 characters to limit input to strictly device names, ensuring descriptions and notes are kept in the correct columns. The MAC address has been set to 17 characters to ensure integrity. Customer_id foreign key to ensure data normalization.

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Customer

```
+-----+
| customer | CREATE TABLE `customer` (
| `customer_id` int NOT NULL,
| `customer_name` varchar(100) DEFAULT NULL,
| `customer_email` varchar(100) DEFAULT NULL,
| `customer_phone_number` varchar(20) DEFAULT NULL,
| PRIMARY KEY (`customer_id`),
| UNIQUE KEY `customer_email` (`customer_email`),
| UNIQUE KEY `customer_phone_number` (`customer_phone_number`)
| ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+
1 row in set (0.00 sec)
```

VARCHAR() has been used similarly in this column to ensure data integrity, as used in other tables. Customer_name and email have been set to 100 characters to ensure limits on data inside the table and ensure correct use of the table. Phone_number has been set to 20 to ensure numbers can be a total of 15 digits to comply with E.164 telecommunication standards, plus room for symbols and leading numbers. Primary key customer_id has been added to ensure customer information is normalized and not redundantly collected. 2 unique constraints have been placed on customer emails and customer phone numbers to ensure data integrity.

Department

```
+-----+
| Table      | Create Table
+-----+
| department | CREATE TABLE `department` (
| `department_id` int NOT NULL,
| `department_name` varchar(100) DEFAULT NULL,
| PRIMARY KEY (`department_id`),
| UNIQUE KEY `department_name` (`department_name`)
| ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+
1 row in set (0.00 sec)
```

Department has been implemented as designed, with a VARCHAR(100) limit on the department_name to ensure storage and integrity. The unique key has been used to correctly identify employees to departments and prevent duplicate entries from being added.

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Employee_compensation

[illegible]

This table has had extensive security measures added and data types changed. Compensation_id has been added as the primary key. Tax_id has been pulled from employee_tax_info to ensure sensitive tax info remains separate. Salary, hourly_rate, and bonus have all been assigned the decimal data type with the parameters (10,2) to limit the length of the number and decimal places to match specific dollar amounts. Employee_id has been pulled from employee to separate contact info and payment info. Three constraint checks have also been added to make sure values for salaries, bonuses, and hourly rates are either equal to or above 0.

Employee

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```
-----+
| employee | CREATE TABLE `employee` (
| `employee_id` int NOT NULL,
| `department_id` int DEFAULT NULL,
| `admin` varchar(50) DEFAULT NULL,
| `first_name` varchar(50) DEFAULT NULL,
| `last_name` varchar(50) DEFAULT NULL,
| `email` varchar(100) DEFAULT NULL,
| `phone_number` varchar(20) DEFAULT NULL,
| PRIMARY KEY (`employee_id`),
| UNIQUE KEY `email` (`email`),
| UNIQUE KEY `phone_number` (`phone_number`),
| KEY `department_id` (`department_id`),
| CONSTRAINT `employee_ibfk_1` FOREIGN KEY (`department_id`) REFERENCES `department` (`department_id`)
| ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+
```

Employee_id has been added to this table as a primary key so basic employee info can be referenced without revealing sensitive data. Department_id has been pulled from department to reference what department they are employed in. Admin, first_name, and last_name have been limited to 50 characters for data integrity and correct input. Email has been limited to 100 characters. Phone number has been limited to 20 characters to ensure compliance with E.164, like in the customer phone number. Email and phone_number have also been set to unique to ensure no duplicates are entered and to ensure data integrity.

Employee_tax_info

```
-----+
| employee_tax_info | CREATE TABLE `employee_tax_info` (
| `tax_id` int NOT NULL,
| `employee_id` int DEFAULT NULL,
| `tax_info` text,
| PRIMARY KEY (`tax_id`),
| KEY `employee_id` (`employee_id`),
| CONSTRAINT `employee_tax_info_ibfk_1` FOREIGN KEY (`employee_id`) REFERENCES `employee` (`employee_id`)
| ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+
```

Employee_tax_info was implemented as planned with the foreign employee_id key and using the tax_id primary key.

Technician_assignments

```
-----+
| technician_assignment | CREATE TABLE `technician_assignment` (
| `assignment_id` int NOT NULL,
| `ticket_id` int DEFAULT NULL,
| `employee_id` int DEFAULT NULL,
| `assigned_date_time` datetime DEFAULT NULL,
| PRIMARY KEY (`assignment_id`),
| KEY `ticket_id` (`ticket_id`),
| KEY `employee_id` (`employee_id`),
| CONSTRAINT `technician_assignment_ibfk_1` FOREIGN KEY (`ticket_id`) REFERENCES `tickets` (`ticket_id`),
| CONSTRAINT `technician_assignment_ibfk_2` FOREIGN KEY (`employee_id`) REFERENCES `employee` (`employee_id`)
| ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+
```

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Assignment_id is used as the primary key in this table. Ticket_id is used as a foreign key from the tickets table. Employee_id is used as a foreign key to identify the employee assigned to the ticket.

Tickets

```
--+
| tickets | CREATE TABLE `tickets` (
  `ticket_id` int NOT NULL,
  `status` varchar(50) DEFAULT NULL,
  `priority` varchar(50) DEFAULT NULL,
  `internal_notes` text,
  `description` text,
  `customer_id` int DEFAULT NULL,
  PRIMARY KEY (`ticket_id`),
  KEY `customer_id` (`customer_id`),
  CONSTRAINT `tickets_ibfk_1` FOREIGN KEY (`customer_id`) REFERENCES `customer` (`customer_id`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+
1 row in set (0.01 sec)
```

Ticket_id is used as the primary key in this table. Status and priority have been limited to 50 characters with VARCHAR(50). Customer_id is a foreign key pulling info from customer to limit customer info visibility.

Ticket_junction_table

```
--+
| ticket_junction_table | CREATE TABLE `ticket_junction_table` (
  `ticket_id` int NOT NULL,
  `device_id` int NOT NULL,
  PRIMARY KEY (`ticket_id`,`device_id`),
  KEY `device_id` (`device_id`),
  CONSTRAINT `ticket_junction_table_ibfk_1` FOREIGN KEY (`ticket_id`) REFERENCES `tickets` (`ticket_id`),
  CONSTRAINT `ticket_junction_table_ibfk_2` FOREIGN KEY (`device_id`) REFERENCES `customer_devices` (`device_id`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+
```

The ticket junction table was needed due to tickets and devices having a many to many relationship. One ticket can relate to many devices, as well as vice versa. This warranted a junction table to be developed.

Representative dataset

The representative dataset that was developed last week in lab2A was loaded into the database. However, only the specified data was added, with the exception of the employee table, to verify database functionality. This led to multiple tables being empty, which have been documented and verified.

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Billing

```
| ticket_junction_table |
| tickets |
+-----+
10 rows in set (0.01 sec)

mysql> DESCRIBE ngts_security_lab;
ERROR 1146 (42S02): Table 'ngts_security_lab.ngts_security_lab' doesn't exist
mysql> DESCRIBE billing;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| billing_id | int | NO | PRI | NULL | |
| customer_id | int | YES | UNI | NULL | |
| customer_billing_address | varchar(255) | YES | | NULL | |
| cardlast_four | varchar(4) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.05 sec)

mysql> 
```

Customer

```
mysql> SELECT * FROM customer;
+-----+-----+-----+-----+
| customer_id | customer_name | customer_email | customer_phone_number |
+-----+-----+-----+-----+
| 1 | North Georgia Carpentry | ngcarpentry@gmail.com | 665-876-9009 |
| 2 | Alabama Electric Company | alabamaemc@aemc.com | 653-788-9888 |
| 3 | John Evans Pest Control | Evans_Pest_Control@gmail.com | 334-664-2233 |
| 4 | North Florida Tech College | northfloridatech@nftech.edu | 406-800-1227 |
| 5 | Superstar Marketing | superstarmarketing@superm.com | 909-903-2256 |
+-----+-----+-----+-----+
5 rows in set (0.01 sec)
```

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Customer devices

```
mysql> SELECT * FROM customer_devices;
```

device_id	customer_id	ip	device_type	host_name	mac_address
1	1	NULL	Printer	NULL	NULL
2	2	NULL	Company Employee Computer	NULL	NULL
3	3	NULL	Company Employee Computer	NULL	NULL
4	4	NULL	Server	NULL	NULL
5	5	NULL	Company Phone	NULL	NULL

5 rows in set (0.00 sec)

Department

```
+-----+-----+
| department_id | department_name |
+-----+-----+
|          1   | Technician      |
|          2   | HR              |
|          3   | Marketing       |
|          4   | Management      |
+-----+-----+
```

4 rows in set (0.01 sec)

Employee

```
mysql> SELECT * FROM employee;
```

employee_id	department_id	admin	first_name	last_name	email	phone_number
1	NULL	NULL	John	Smith	NULL	NULL
2	NULL	NULL	Lisa	Bernard	NULL	NULL
3	NULL	NULL	Cam	Heshar	NULL	NULL
4	NULL	NULL	Bo	Nicholas	NULL	NULL
5	NULL	NULL	Gunnar	Stapleton	NULL	NULL
6	NULL	NULL	Kirby	Lethront	NULL	NULL

6 rows in set (0.00 sec)

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Technician assignments

```
mysql> DESCRIBE technician_assignment;
+-----+-----+-----+-----+-----+-----+
| Field          | Type   | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| assignment_id  | int    | NO   | PRI | NULL    |       |
| ticket_id      | int    | YES  | MUL | NULL    |       |
| employee_id    | int    | YES  | MUL | NULL    |       |
| assigned_date_time | datetime | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.02 sec)

mysql>
```

Tickets

```
mysql> SELECT * FROM tickets;
+-----+-----+-----+-----+-----+-----+
| ticket_id | status | priority | internal_notes | description | customer_id |
+-----+-----+-----+-----+-----+-----+
| 1 | NULL | NULL | NULL | Wont respond to pings | 4 |
| 2 | NULL | NULL | NULL | Needs MDM installed | 5 |
| 3 | NULL | NULL | NULL | Printer won't connect to computers over wifi | 1 |
| 4 | NULL | NULL | NULL | Gets stuck and won't print if job is queued | 1 |
| 5 | NULL | NULL | NULL | Needs antivirus installed | 2 |
| 6 | NULL | NULL | NULL | Needs EDR installed | 2 |
| 7 | NULL | NULL | NULL | Needs new wifi configured | 2 |
| 8 | NULL | NULL | NULL | Needs data migration | 3 |
| 9 | NULL | NULL | NULL | Needs new OS install | 3 |
+-----+-----+-----+-----+-----+-----+
9 rows in set (0.00 sec)
```

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Ticket junction table

```
mysql> SELECT * FROM ticket_junction_table;
+-----+-----+
| ticket_id | device_id |
+-----+-----+
|          3 |          1 |
|          4 |          1 |
|          5 |          2 |
|          6 |          2 |
|          7 |          2 |
|          8 |          3 |
|          9 |          3 |
|          1 |          4 |
|          2 |          5 |
+-----+-----+
9 rows in set (0.00 sec)
```

Empty Sets(billing, employee_tax_info, employee_compensation, technician_assignment)

```
mysql> SELECT * FROM billing;
Empty set (0.00 sec)

mysql> SELECT * FROM employee_tax_info;
Empty set (0.00 sec)

mysql> SELECT * FROM employee_compensation;
Empty set (0.00 sec)

mysql> SELECT * FROM technician_assignment;
Empty set (0.00 sec)
```

Successful Tests

Two different successful tests were performed and documented to verify functionality. One adding employee info and one adding customer info.

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```
mysql> INSERT INTO employee (employee_id, first_name, last_name)
-> VALUES
-> (1, 'John', 'Smith'),
-> (2, 'Lisa', 'Bernard'),
-> (3, 'Cam', 'Heshar'),
-> (4, 'Bo', 'Nicholas'),
-> (5, 'Gunnar', 'Stapleton'),
-> (6, 'Kirby', 'Lethront');
Query OK, 6 rows affected (0.02 sec)
Records: 6 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO customer (customer_id, customer_name, customer_email, customer_phone_number)
-> VALUES
-> (1, 'North Georgia Carpentry', 'ngcarpentry@gmail.com', '665-876-9009'),
-> (2, 'Alabama Electric Company', 'alabamaemc@aemc.com', '653-788-9888'),
-> (3, 'John Evans Pest Control', 'Evans_Pest_Control@gmail.com', '334-664-2233'),
-> (4, 'North Florida Tech College', 'northfloridatech@nftech.edu', '406-800-1227'),
-> (5, 'Superstar Marketing', 'superstarmarketing@superm.com', '909-903-2256');
Query OK, 5 rows affected (0.02 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

Intentionally Invalid Tests

Four intentionally invalid tests were performed to verify constraint functionality and key controls.

Card_last_four

```
mysql> INSERT INTO billing (cardlast_four)
-> VALUES
-> ('34456')
-> ;
ERROR 1406 (22001): Data too long for column 'cardlast_four' at row 1
mysql>
```

Card_last_four was tested to verify that data longer than 4 characters would be rejected.

Card_last_four

```
mysql> INSERT INTO billing (cardlast_four, billing_id) VALUES ('344', 2);
ERROR 3819 (HY000): Check constraint 'billing_chk_1' is violated.
mysql>
```

Card_last_four was tested to verify that data shorter than 4 characters would be rejected due to the check = 4 constraint.

Department_name

```
mysql> INSERT INTO department (department_id, department_name) VALUES (6, 'Technician');
ERROR 1062 (23000): Duplicate entry 'Technician' for key 'department.department_name'
mysql>
```

Department_name was verified to confirm that duplicate values would be rejected.

Department_id

```
mysql> INSERT INTO department ( department_id, department_name) VALUES (1, 'accounting');
ERROR 1062 (23000): Duplicate entry '1' for key 'department.PRIMARY'
mysql>
```

Department_id was tested to verify that duplicate ID values would be rejected.

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Reflection

These errors are not application errors because they are designed and implemented to happen. These aren't the result of bad syntax or misbehaving software, they are the result of purposeful design for specific circumstances set by the database admin.

Peer Review

For this peer review, I was able to get peer reviewed by someone in my family in the security industry. The main problems that I was having were related to my constraints and restrictive character limits. With his insight, I was able to more accurately limit input to the table and set more specific constraints, like my check constraint for my card_last_four.

Week 5 Snapshot

